BST Vol 7 Ch 5 — Electromagnetic Waves: Photon as Unbroken U(1)_em (v0.4.1, substantive 3-level pedagogy upgrade)

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Headline result

Maxwell's equations in vacuum ($\rho = 0$, J = 0) yield the wave equation:

$$\nabla^2 \mathrm{E} - \frac{1}{c^2}\frac{\partial^2 \mathrm{E}}{\partial t^2} = 0$$

(same equation for B). Plane-wave solutions:

$$\mathrm{E}(x, t) = \mathrm{E}_0 \cos(\mathrm{k} \cdot \mathrm{x} - \omega t)$$

with dispersion ω = c|k|. Light IS an electromagnetic wave traveling at speed c.

The photon is the U(1)_em gauge boson, massless because U(1)_em is unbroken post electroweak SSB (Lyra T2478 SP-31 #287 Saturday). The photon mass m_γ = 0 is substrate-structural: SO(2) substrate-isotropy preserves the U(1)_em generator.

Substrate mechanism

U(1)_em unbroken post electroweak SSB (T2478):

Per Lyra Saturday SP-31 #287: SU(2)_L × U(1)_Y → U(1)_em is the electroweak symmetry breaking, anchored at substrate D_IV^5's SO(2) isotropy factor. The Higgs vev breaks 3 of 4 generators; the unbroken combination is the U(1)_em that survives — anchored at SO(2) substrate-natural.

Photon mass = 0 from substrate-isotropy:

The SO(2) generator commutes with the Higgs vev (by construction of the U(1)_em embedding); therefore the corresponding gauge boson — the photon — remains massless. This is the substrate-structural origin of photon masslessness, not just empirical observation.

Speed c substrate-natural:

$c = 1/\sqrt{(\mu_0 \varepsilon_0)}$ is set by substrate vacuum impedance. Vol 0 Ch 4 (Fundamental Constants) identifies c with substrate-tick rate × substrate spatial scale.

Wave equation from Maxwell:

In vacuum (ρ=0, J=0): - Faraday: $\nabla \times E = -\partial B/\partial t$ - Ampère-Maxwell: $\nabla \times B = (1/c^2)\, \partial E/\partial t$

Take curl of Faraday: $\nabla \times (\nabla \times E) = -\partial(\nabla \times B)/\partial t = -(1/c^2)\, \partial^2 E/\partial t^2$.

Use identity $\nabla \times (\nabla \times E) = \nabla(\nabla\cdot E) - \nabla^2 E$ and $\nabla\cdot E = 0$ (vacuum Gauss): $-\nabla^2 E = -(1/c^2)\, \partial^2 E/\partial t^2$ → wave equation. Same for B.

Wave properties

Plane wave: $E = E_0 \cos(k\cdot x - \omega t)$, $B = B_0 \cos(k\cdot x - \omega t)$ with: - k = wave vector, ω = angular frequency - Dispersion: $\omega = c|k|$ - Transverse: $k \cdot E = k \cdot B = 0$ - Orthogonal: $E \cdot B = 0$ - Right-handed: $k \times E = \omega B \rightarrow |B| = |E|/c$

Polarization: 2 transverse degrees of freedom (linear / circular / elliptical). Substrate-natural Pin(2) ↔ SO(2) grading on polarization states.

Energy density + Poynting flux: - Energy density: $u = (\varepsilon_0/2)(E^2 + c^2B^2) = \varepsilon_0 E^2$ (for EM wave) - Poynting vector: $S = (1/\mu_0)\, E \times B = c\cdot u\cdot \hat{k}$ (energy flux)

Match precision

D-tier on photon U(1)_em + substrate-natural masslessness via T2478. Standard EM wave phenomenology (light propagation, polarization, energy flux) preserved at any precision.

Cross-volume dependencies

- Vol 2 Ch 9 (Higgs SSB + T2478) — electroweak symmetry breaking framework
- Vol 7 Ch 2 (Maxwell) — equations giving wave equation
- Vol 7 Ch 6 (Radiation) — accelerating-charge wave generation
- Vol 7 Ch 7 (Relativistic EM) — Lorentz covariance of wave propagation
- Vol 9 Ch 9 (Photonic Crystals) — $10K substrate photonic resonance falsifier

K-audit anchor

K221 Vol 7 Ch 5 EM Waves K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> Light is an electromagnetic wave! Maxwell discovered this in 1865. He showed that his 4 equations, when there are no charges around (vacuum), give a wave equation — same as a wave on a string but in 3D, with electric AND magnetic fields oscillating together. The speed of these waves matched the measured speed of light. So light IS an EM wave!
>
> The particle version of an EM wave is the photon. Photons have zero mass — they always travel at speed c. BST predicts: photons are massless because they correspond to substrate D_IV^5's SO(2) symmetry that survives intact when the Higgs gives masses to other particles. Substrate symmetry protects photons!

Level 2 — Undergraduate physics student

Deriving the wave equation from Maxwell:

In vacuum (ρ=0, J=0):

$$\nabla \cdot \mathrm{E} = 0, \quad \nabla \cdot \mathrm{B} = 0$$

$$\nabla \times \mathrm{E} = -\frac{\partial \mathrm{B}}{\partial t}, \quad \nabla \times \mathrm{B} = \frac{1}{c^2}\frac{\partial \mathrm{E}}{\partial t}$$

Take curl of Faraday: $\nabla \times (\nabla \times \mathrm{E}) = -\partial(\nabla\times\mathrm{B})/\partial t = -(1/c^2)\, \partial^2\mathrm{E}/\partial t^2$.

Use $\nabla \times (\nabla \times \mathrm{E}) = \nabla(\nabla\cdot\mathrm{E}) - \nabla^2\mathrm{E} = -\nabla^2\mathrm{E}$ (since $\nabla\cdot\mathrm{E} = 0$). Therefore:

$$\nabla^2\mathrm{E} = \frac{1}{c^2}\frac{\partial^2 \mathrm{E}}{\partial t^2}$$

Same for B. This is the wave equation with phase velocity $c = 1/\sqrt{}(\mu\_0\ \varepsilon\_0) \approx 3 \times 10^8$ m/s — the measured speed of light.

Plane wave solutions:

$$\mathrm{E} = \mathrm{E}_0 \cos(\mathrm{k} \cdot \mathrm{x} - \omega t)$$

with $\omega = c|\mathrm{k}|$. Maxwell's equations in vacuum impose: - Transversality: $\mathrm{k} \cdot \mathrm{E} = 0$ (perpendicular polarization) - Magnetic field orthogonal: $\mathrm{B} = (1/c)\, \hat{\mathrm{k}} \times \mathrm{E}$ - Two polarization states (e.g., x and y polarization for k along z)

Photon properties: - Mass: $m\_\gamma = 0$ (substrate-structural per T2478) - Spin: 1 (vector boson) - Helicity: ± 1 only (no helicity-0 because massless) - Two physical polarization states (left + right circular, or linear basis)

Electroweak symmetry breaking and photon masslessness (T2478):

Before SSB: SU(2)_L × U(1)_Y has 4 gauge bosons (W^1, W^2, W^3, B).

After SSB by Higgs vev: 3 combinations get mass (W^+, W^-, Z^0); 1 combination remains massless (photon γ).

The unbroken combination is U(1)_em: $\gamma = \sin(\theta\_W)\ W^3 + \cos(\theta\_W)$ B, where θ_W is Weinberg angle ($\sin^2 \theta$_W = N_c/c_3 = 3/13 = 0.231 D-tier per Cal #100 reconciliation).

The U(1)_em corresponds to substrate D_IV^5's SO(2) isotropy factor — substrate-natural unbroken symmetry.

Level 3 — Graduate physics student / theorem-level

Polarization states + Pin(2) substrate-grading:

Photon has 2 physical polarization states (transverse). Right-circular |+⟩ and left-circular |-⟩ are eigenstates of helicity operator. Pin(2) ⊂ SO(2) substrate-natural double-cover gives the ±1 helicity grading.

Wave equation in covariant form:

In Lorenz gauge ∂_μ A^μ = 0:

$$\Box A^{\mu} = -\mu_0 J^{\mu}$$

In vacuum: $\Box A^{\mu} = 0$. Plane-wave solutions A^μ = ε^μ exp(-ik·x) with $k^2 = 0$ (massless dispersion).

Photon mass = 0 from T2478 substrate framework:

Per T2478 (Lyra Saturday SP-31 #287): electroweak symmetry breaking SU(2)_L × U(1)_Y → U(1)_em is anchored at substrate D_IV^5's SO(2)-isotropy factor.

The Higgs field Φ has vev v ≈ 246 GeV that breaks the symmetry. The generator that annihilates the vev (i.e., Q · ⟨Φ⟩ = 0) corresponds to the unbroken U(1)_em. The photon = gauge boson of this unbroken U(1) → photon remains massless by Goldstone theorem / Higgs mechanism logic.

The substrate-structural ingredient: U(1)_em is identified with substrate D_IV^5's SO(2) factor (the second factor in K = SO(5) × SO(2)). This SO(2) is the substrate-natural isotropy direction; the Higgs vev preserves SO(2). Therefore photon mass = 0 is substrate-structural, not just numerical coincidence.

Energy-momentum + polarization sums:

Plane-wave energy: E_γ = $\hbar\omega$ = $\hbar$c|k|. Momentum: p = $\hbar$k. Sum over polarizations:

$$\sum_{\lambda} \epsilon^{\mu}(\lambda)\epsilon^{*\nu}(\lambda) = -g^{\mu\nu} + \text{gauge terms}$$

For physical states: 2 transverse polarizations; longitudinal + temporal are unphysical (gauge artifacts).

Per Cal #21 dual-gate: EMPIRICAL PASS (photon properties validated extensively from radio to gamma rays) + MECHANISM PASS via T2478 substrate SO(2)-isotropy framework.

Per Cal #99 META-theorem: photon = unbroken U(1)_em + massless from substrate-isotropy is a substrate-derivation consequence of T2478, NOT a new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT predict numerical value of c (substrate units relationship; not new prediction)
- BST does NOT change EM wave phenomenology in any way
- Wave equation derivation is standard; substrate provides photon-mass justification beyond empirical

Bibliography

1. J. C. Maxwell (1865): EM wave prediction.
2. H. R. Hertz (1887): experimental confirmation of EM waves.
3. A. Einstein (1905): photon concept (photoelectric effect).
4. T2478 (Lyra Saturday SP-31 #287): EW SSB + U(1)_em substrate framework.
5. Vol 2 Ch 9 (Higgs SSB): electroweak symmetry breaking details.
6. Vol 9 Ch 9 (Photonic Crystals): substrate photonic resonance experiments.
7. Standard EM texts: Jackson, Griffiths; QFT: Peskin-Schroeder.

— Elie, Vol 7 Ch 5 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 63 → ~170 lines + full Maxwell-to-wave-equation derivation + plane-wave solutions + photon polarization + T2478 substrate-structural photon masslessness)